

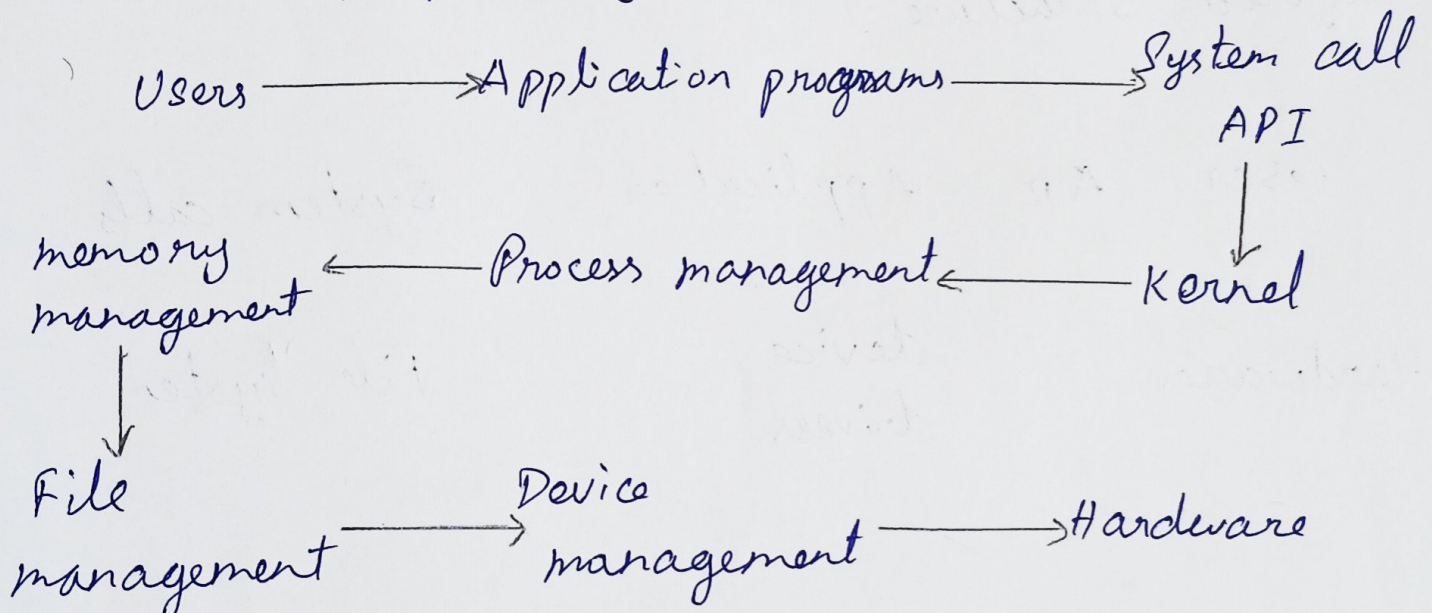
1) a). Type of operating System:

A multi user, multitasking operating system is suitable

Justification:

- \* Supports many users simultaneously
- \* Executes multiple processes at the same time
- \* Shares CPU, memory and I/O efficiently
- \* provides security and protection between users

b). Structure of operating system:





## 1) System call segments

fork ( ) ; // Create processes  
exec ( ) ; // Load new program  
wait ( ) ; // Parent waits  
exit ( ) ; // child terminates

## 2) a). File and device system calls:

### File system calls

- \* open ( )
- \* read ( )
- \* write ( )
- \* close ( )
- \* lseek ( )

### Device system calls

- \* lseek ( )
- \* read ( )
- \* write ( )
- \* open ( )
- \* close ( )

## 2) Layered structure :

